Algebra & Graphs

ClipSAT Printable Practice Suite (clipsat.org) • Topic 2.01

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A bag contains 6 green counters and 4 yellow counters. A counter is chosen at random, its colour recorded, and it is REPLACED. A second counter is then chosen. What is the probability that both counters are the same colour?

Since the counter is replaced, events are independent: $P(\text{Green, Green}) = \left(\frac{6}{10}\right)^2 = \frac{36}{100}$ $P(\text{Yellow, Yellow}) = \left(\frac{4}{10}\right)^2 = \frac{16}{100}$ $P(\text{Same Colour}) = \frac{36}{100} + \frac{16}{100} = \frac{52}{100} = \frac{13}{25}$.

2 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A bank account pays compound interest of 3% per year. How many complete years will it take for an initial deposit of USD 2000 to exceed USD 3000?

We need $2000(1.03)^n > 3000 \Rightarrow (1.03)^n > 1.5$. Taking logarithms: $n\ln(1.03) > \ln(1.5) \Rightarrow n > \frac{\ln(1.5)}{\ln(1.03)} \approx \frac{0.405465}{0.029559} \approx 13.72$. The minimum number of complete years is 14. [M1 for setting up inequality, A1 for 14 years]

3 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Port B is 40 km from Port A on a bearing of 120°. Port C is 50 km from Port A on a bearing of 030°. Calculate the distance between Port B and Port C.

The angle between the two paths at Port A is $120^\circ - 030^\circ = 90^\circ$. Since this is a right triangle, use Pythagoras: $BC = \sqrt{40^2 + 50^2} = \sqrt{1600 + 2500} = \sqrt{4100} \approx 64.03$ km ≈ 64.0 km. [M1 for angle at A = 90, M1 for Pythagoras, A1 for 64.0]

4 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

A drawer contains 5 blue socks, 4 grey socks, and 3 black socks. Two socks are pulled at random without replacement. What is the probability of pulling two socks of different colours?

Total socks = 12. Total pairs = $\binom{12}{2} = 66$. Number of same-colour pairs: $\binom{5}{2} + \binom{4}{2} + \binom{3}{2} = 10 + 6 + 3 = 19$. Therefore, the probability that the two socks have different colours is: $P(\text{Different}) = 1 - \frac{19}{66} = \frac{47}{66}$.

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5 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In a circle of radius 10 cm, an arc subtends a central angle of 72°. Calculate the perimeter of this sector to 1 decimal place.

Arc length: $s = \frac{72}{360} \times 2\pi(10) = \frac{1}{5} \times 20\pi = 4\pi \approx 12.57$ cm. Perimeter of sector = arc + 2 radii = $12.57 + 2(10) = 32.57 \approx 32.6$ cm. [M1 for arc length, M1 for adding 2r, A1 for 32.6]

6 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A vertical pole of height 12 m casts a shadow of length 9 m on horizontal ground. Calculate the angle of elevation of the sun to the nearest degree.

Use tangent: $\tan\theta = \frac{\text{opposite}}{\text{adjacent}} = \frac{12}{9} \approx 1.333 \Rightarrow \theta = \tan^{-1}(1.333) \approx 53.1^\circ \approx 53^\circ$. [M1 for tan(theta)=12/9, A1 for 53]

7 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

For the model $f(x) = 6x + 3$ in adding and subtracting fractions, evaluate $f(5)$.

Final Answer: 33. Substitute $x = 5$: $f(5) = 6(5) + 3 = 33$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

For the model $f(x) = 2x + 4$ in fractions of a quantity, evaluate $f(2)$.

Final Answer: 8. Substitute $x = 2$: $f(2) = 2(2) + 4 = 8$.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A data set recorded for fractions of a quantity consists of the four values $\{16, 20, 24, 30\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 22.5. The sum is $16 + 20 + 24 + 30 = 90$, and dividing by 4 gives 22.5.

10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Solve the quadratic equation $2x^2 + 5x - 4 = 0$. Give your solutions correct to 2 decimal places.

Identify coefficients: $a = 2$, $b = 5$, $c = -4$. $x = \frac{-5 \pm \sqrt{5^2 - 4(2)(-4)}}{2(2)} = \frac{-5 \pm \sqrt{25 + 32}}{4} = \frac{-5 \pm \sqrt{57}}{4}$ $x_1 = \frac{-5 + 7.5498}{4} = \frac{2.5498}{4} \approx 0.64$ $x_2 = \frac{-5 - 7.5498}{4} = \frac{-12.5498}{4} \approx -3.14$ Solutions: $x = 0.64$ or $x = -3.14$ (both to 2 d.p.).

11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

[Algebra & Graphs · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

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12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

The nth term of a sequence is $3n - 7$. (a) Find the 10th term. (b) Which term has value 23?

(a) $3(10) - 7 = 23$. (b) $3n - 7 = 23 \Rightarrow 3n = 30 \Rightarrow n = 10$. Term 10 has value 23.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the equation of the perpendicular bisector of the line segment joining $P(2, -1)$ and $Q(8, 7)$. Give your answer in the form $y = mx + c$.

Step 1: Find the midpoint M of PQ: $M = \left(\frac{2+8}{2}, \frac{-1+7}{2}\right) = (5, 3)$ Step 2: Find the gradient of line PQ: $m_{PQ} = \frac{7-(-1)}{8-2} = \frac{8}{6} = \frac{4}{3}$ Step 3: Determine the perpendicular gradient: $m_{\perp} = -\frac{3}{4}$ Step 4: Form the equation passing through $M(5, 3)$: $y - 3 = -\frac{3}{4}(x - 5) \Rightarrow y - 3 = -\frac{3}{4}x + \frac{15}{4}$
$y = -\frac{3}{4}x + \frac{27}{4}$ or $y = -0.75x + 6.75$

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

In triangle OAB, $\overrightarrow{OA} = 3\mathbf{a}$ and $\overrightarrow{OB} = 6\mathbf{b}$. Point P lies on AB such that $AP : PB = 1 : 2$. Find $\overrightarrow{OP}$ in terms of $\mathbf{a}$ and $\mathbf{b}$ in its simplest form.

Step 1: Express $\overrightarrow{AB}$: $\overrightarrow{AB} = \overrightarrow{AO} + \overrightarrow{OB} = -3\mathbf{a} + 6\mathbf{b}$ Step 2: Find $\overrightarrow{AP}$: $\overrightarrow{AP} = \frac{1}{3}\overrightarrow{AB} = \frac{1}{3}(-3\mathbf{a} + 6\mathbf{b}) = -\mathbf{a} + 2\mathbf{b}$ Step 3: Find $\overrightarrow{OP}$: $\overrightarrow{OP} = \overrightarrow{OA} + \overrightarrow{AP} = 3\mathbf{a} + (-\mathbf{a} + 2\mathbf{b}) = 2\mathbf{a} + 2\mathbf{b}$

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 8.01 Probability & Statistics

A solid toy is formed by attaching a cone of radius 4 cm and height 9 cm to the flat face of a solid hemisphere of radius 4 cm. Calculate the total volume of the toy. Give your answer correct to 3 significant figures.

Step 1: Volume of cone: $V_{\text{cone}} = \frac{1}{3}\pi r^2 h = \frac{1}{3}\pi(4^2)(9) = 48\pi \text{ cm}^3$ Step 2: Volume of hemisphere: $V_{\text{hemi}} = \frac{1}{2} \times \frac{4}{3}\pi r^3 = \frac{2}{3}\pi(4^3) = \frac{128}{3}\pi \text{ cm}^3$ Step 3: Total volume:

$V_{\text{total}} = 48\pi + \frac{128}{3}\pi = \frac{272}{3}\pi \approx 284.838\ldots \approx 285 \text{ cm}^3$ (3 s. f.)

16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 9.01 Functions

In triangle XYZ, $XY = 7$ cm, $YZ = 10$ cm, and angle $XYZ = 65^\circ$. Calculate (a) the length of XZ to 3 significant figures and (b) the area of triangle XYZ.

(a) Use the Cosine Rule: $XZ^2 = 7^2 + 10^2 - 2(7)(10)\cos(65^\circ) = 49 + 100 - 140(0.4226) = 149 - 59.166 = 89.834$
$XZ = \sqrt{89.834} \approx 9.4779\ldots \approx 9.48$ cm (3 s. f.) (b) Area of triangle:

$\text{Area} = \frac{1}{2}ab\sin C = \frac{1}{2}(7)(10)\sin(65^\circ) = 35(0.9063) \approx 31.7 \text{ cm}^2$ (3 s. f.)

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17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 1.01 Number

A car travels a distance of $d = 180$ km correct to the nearest 10 km in a time of $t = 2.4$ hours correct to 1 decimal place. Calculate the upper bound for the average speed of the car in km/h.

Find the bounds for distance and time: $d_{\text{upper}} = 180 + 5 = 185$ km $t_{\text{lower}} = 2.4 - 0.05 = 2.35$ hours To maximize speed $s = \frac{d}{t}$, divide the upper distance by the lower time:
$s_{\text{upper}} = \frac{185}{2.35} \approx 78.723\ldots \approx 78.7$ km/h (3 s. f.)

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 10.01 Sets & Venn Diagrams

The table shows the time, t minutes, taken by 50 students to complete a homework task. Intervals: $0 < t \leq 10$ (freq: 8), $10 < t \leq 20$ (freq: 14), $20 < t \leq 30$ (freq: 18), $30 < t \leq 40$ (freq: 10). Calculate an estimate of the mean time taken.

Find class midpoints (x) and products (fx):Midpoint 5: $8 \times 5 = 40$Midpoint 15: $14 \times 15 = 210$Midpoint 25: $18 \times 25 = 450$Midpoint 35: $10 \times 35 = 350$ $\sum fx = 40 + 210 + 450 + 350 = 1050$
Estimated Mean $= \frac{\sum fx}{\sum f} = \frac{1050}{50} = 21$ minutes

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 3.01 Coordinate Geometry

In a survey of 60 students, 35 study French, 28 study German, and 12 study both languages. Find (a) the number of students who study neither language, and (b) the probability that a student chosen at random studies French given that they study German.

(a) Inclusion-Exclusion: $n(F \cup G) = n(F) + n(G) - n(F \cap G) = 35 + 28 - 12 = 51$
$n(\text{neither}) = 60 - 51 = 9$ students (b) Conditional probability: $P(F \mid G) = \frac{n(F \cap G)}{n(G)} = \frac{12}{28} = \frac{3}{7} \approx 0.429$

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 4.01 Mensuration

In a cyclic quadrilateral $ABCD$, angle $ABC = (2x + 15)^\circ$ and angle $ADC = (3x - 10)^\circ$. Find the value of x and hence the size of angle ABC. Give a geometric reason for your equation.

Geometric reason: Opposite angles of a cyclic quadrilateral sum to 180°. $(2x + 15) + (3x - 10) = 180$
$5x + 5 = 180 \Rightarrow 5x = 175 \Rightarrow x = 35$ Calculate angle ABC: $\angle ABC = 2(35) + 15 = 70 + 15 = 85^\circ$

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 4x + 8$ in name, identify, order, and compare fractions, evaluate $f(2)$.

a. 16 ✓

b. 20

c. 8

d. 20

Substitute $x = 2$: $f(2) = 4(2) + 8 = 16$.

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 5x + 9$ in equivalent and simplified fractions, evaluate $f(2)$.

a. 19 ✓

b. 24

c. 10

d. 23

Substitute $x = 2$: $f(2) = 5(2) + 9 = 19$.

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for name, identify, order, and compare fractions consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

a. 17.5

b. 18.5

c. 19.5 ✓

d. 21.5

The sum is $13 + 17 + 21 + 27 = 78$, and dividing by 4 gives 19.5.

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for equivalent and simplified fractions consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

a. 18.5

b. 19.5

c. 20.5 ✓

d. 22.5

The sum is $14 + 18 + 22 + 28 = 82$, and dividing by 4 gives 20.5.

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25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A data set recorded for adding and subtracting fractions consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean?

a. 19.5

b. 20.5

c. 21.5 ✓

d. 23.5

The sum is $15 + 19 + 23 + 29 = 86$, and dividing by 4 gives 21.5.

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A data set recorded for multiplying fractions consists of the four values $\{17, 21, 25, 31\}$. What is the arithmetic mean?

a. 21.5

b. 22.5

c. 23.5 ✓

d. 25.5

The sum is $17 + 21 + 25 + 31 = 94$, and dividing by 4 gives 23.5.

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For the model $f(x) = 3x + 5$ in multiplying fractions, evaluate $f(3)$.

a. 14 ✓

b. 17

c. 9

d. 18

Substitute $x = 3$: $f(3) = 3(3) + 5 = 14$.

28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A data set recorded for fractions and division consists of the four values $\{18, 22, 26, 32\}$. What is the arithmetic mean?

a. 22.5

b. 23.5

c. 24.5 ✓

d. 26.5

The sum is $18 + 22 + 26 + 32 = 98$, and dividing by 4 gives 24.5.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For the model $f(x) = 4x + 6$ in fractions and division, evaluate $f(3)$.

a. 18 ✓

b. 22

c. 12

d. 22

Substitute $x = 3$: $f(3) = 4(3) + 6 = 18$.

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

The diagram shows a curve with equation $y = ax^2 + bx + c$ and a straight line with equation $y = mx + d$. The curve and the line intersect at points A and B. The vertex of the parabola is at point V. The x-coordinates of the points A, B, and V are given as $x_A = -1$, $x_B = 3$, and $x_V = 1$ respectively. The y-coordinate of the vertex V is $y_V = -4$. Determine the value of a.

a. **1** ✓

b. 2

c. -1

d. -2

The vertex of a parabola $y = ax^2 + bx + c$ is located at $x = -\frac{b}{2a}$. We are given that the x-coordinate of the vertex V is $x_V = 1$. Therefore, $1 = -\frac{b}{2a}$, which implies $b = -2a$. The parabola passes through its vertex V at $(1, -4)$. Substituting these coordinates into the parabola equation: $-4 = a(1)^2 + b(1) + c$ $-4 = a + b + c$. Substituting $b = -2a$: $-4 = a + (-2a) + c$ $-4 = -a + c$ (Equation 1) The parabola and the line intersect at points A and B, with x-coordinates $x_A = -1$ and.

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

The equation $y = 2x^2 - 8$ crosses the x-axis at:

a. $x = \pm 2$ ✓

b. $x = \pm 4$

c. $x = 2$

d. $x = \pm\sqrt{8}$

Applying Algebra & Graphs principles, $2x^2 = 8 \Rightarrow x = \pm 2$ Therefore, the verified correct option is $x = \pm 2$.

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32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Solve the simultaneous equations $x + y = 7, x - y = 3$.

a. $x = 4, y = 3$

b. $x = 5, y = 2$ ✓

c. $x = 3, y = 4$

d. $x = 2, y = 5$

Applying Algebra & Graphs principles, Add: $2x = 10 \Rightarrow x = 5, y = 2$. Therefore, the verified correct option is $x = 5, y = 2$.

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Solve $2x^2 + 6x - 20 = 0$ using the quadratic formula. Give the larger root, as a fraction if needed.

a. -5

b. **2** ✓

c. 1

d. 3

$x = \frac{-6 \pm 14}{4} \Rightarrow x = -5$ or 2. Larger root: 2.

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the nth term of the sequence $2, 6, 10, 14, \ldots$

a. $4n$

b. $4n - 4$

c. $5n - 2$

d. $4n - 2$ ✓

Common difference $= 4$. nth term $= 4n - 2$.

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Solve the simultaneous equations: $x + y = 9, x - y = -5$. Find x.

a. 9

b. **2** ✓

c. -6

d. 0

Elimination gives $x = 2$ (and $y = 7$).

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36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A right triangular prism has a cross-section of a right triangle with legs 5 cm and 12 cm. The length of the prism is 20 cm. What is its volume?

a. **600 cm³** ✓

b. 1200 cm³

c. 300 cm³

d. 450 cm³

Cross-section area = $\frac{1}{2} \times 5 \times 12 = 30$ cm^2. Volume = $30 \times 20 = 600$ cm^3. [M1 A1]

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A triangular prism has a cross-sectional (triangle) area of 24 cm^2 and length 4 cm. Find its volume.

a. 99 cm^3

b. 98 cm^3

c. **96 cm^3** ✓

d. 97 cm^3

Volume $=$ cross $-$ sectional area $\times$ length $= 24 \times 4 = 96$ cm^3.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 10.01 Sets & Venn Diagrams

A function is defined by $f(x) = x^2 - 6x + 5$ for $x \geq 3$. \\ The graph of $y = f(x)$ is a parabola. A straight line with equation $y = k$ intersects the graph of $y = f(x)$ at exactly one point. \\ Find the possible value(s) of k.

a. $k = -4$

b. $k = 5$

c. $k = -4$ or $k = 5$ ✓

d. $k \geq 5$

The function is given by $f(x) = x^2 - 6x + 5$ for $x \geq 3$. \\ To find the vertex of the parabola, we can complete the square: $f(x) = (x^2 - 6x + 9) - 9 + 5 = (x - 3)^2 - 4$. \\ The vertex of the parabola is at $(3, -4)$. \\ Since the domain is restricted to $x \geq 3$, the graph of $y = f(x)$ starts at its vertex and extends upwards as x increases. \\ A horizontal line $y = k$ intersects the graph of $y = f(x)$ at exactly one point if: \\ 1. The line passes through the vertex. In this case, $k = -4$. \\ 2. The.

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39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 3.01 Coordinate Geometry

In triangle ABC, angle ABC = 90°, AB = 10 cm, and angle BAC = θ. Given that $\sin\theta = \frac{3}{5}$, find the length of AC.

a. 16.7 cm

b. 8.0 cm

c. **12.5 cm** ✓

d. 6.0 cm

We are given a right-angled triangle ABC, where angle ABC = 90°. We know that AB = 10 cm and angle BAC = θ. We are also given that $\sin\theta = \frac{3}{5}$. In a right-angled triangle, the sine of an angle is defined as the ratio of the length of the side opposite the angle to the length of the hypotenuse. Therefore, $\sin\theta = \frac{\text{Opposite}}{\text{Hypotenuse}}$. In triangle ABC, the side opposite to angle θ (angle BAC) is BC, and the hypotenuse is AC. So.

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 4.01 Mensuration

On a histogram, a class interval is $20 \le x < 30$ with a frequency of 25. What is the frequency density for this bar?

a. **2.50** ✓

b. 5.00

c. 25

d. 10

Class width = $30 - 20 = 10$. Frequency density = $\frac{\text{Frequency}}{\text{Class Width}} = \frac{25}{10} = 2.50$. [M1 A1]

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 5.01 Geometry

A circle has radius 6 cm. Calculate the area of a sector with central angle 60° in terms of π.

a. 6π cm^2 ✓

b. 12π cm^2

c. 3.0π cm^2

d. 360π cm^2

Formula: $A = \frac{\theta}{360} \times \pi r^2 = \frac{60}{360} \times \pi(6)^2 = 6\pi$ cm^2. [M1 A1]

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42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 6.01 Trigonometry

Find the range of the data set: $3, 5, 12, 14, 15$.

a. 18

b. 7

c. **12** ✓

d. 6

Applying Probability & Statistics principles, Range $= 15 - 3 = 12$. Therefore, the verified correct option is 12.

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 7.01 Transformations & Vectors

Volume of a cone with radius 3 and height 4:

a. 12π ✓

b. 16π

c. 36π

d. 48π

Applying Geometry principles, $V = \frac{1}{3}\pi(9)(4) = 12\pi$ Therefore, the verified correct option is 12π.

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 8.01 Probability & Statistics

Interior angle of a regular pentagon:

a. 100°

b. **108°** ✓

c. 112°

d. 120°

Applying Geometry principles, $\frac{(5-2)\times 180}{5} = 108°$ Therefore, the verified correct option is 108°.

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 9.01 Functions

A circle has radius 17 cm. A chord has length 30 cm. What is the perpendicular distance from the centre of the circle to the chord?

a. **8 cm** ✓

b. 7 cm

c. 9 cm

d. 10 cm

Half chord = 15 cm. Distance = $\sqrt{17^2 - 15^2} = \sqrt{289 - 225} = \sqrt{64} = 8$ cm. [M1 A1]

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